```
%Find the approximate value of t^2y'=y+3t  on interval [1,4] ,initial condition
%y(1)=-2 and plot solution cure
clc
clear all;
f=@(t,y)(y+3*t)/t^2
tspan=[1 4];
y1=-2
[t y]=ode45(f,tspan,y1)
plot(t,y)
```

```
%Find numerical solution using ode45 command if dx/dt=3e^(-t)  time interval
%[0 5] initial condition x(0)=0 and plot cure
clc;
clear all;
clc;
clear all;
f=@(t,x)3*exp(-t);
tspan=[0 5];
x0=0;
[t,x]=ode45(f,tspan,x0)
plot(t,x)
```

```
%Find approximate solution if dy/dt=y/t^2 +2/t
%, y(0)=1 interval [1 5] plot graph
clc;
clear all;
f=@(t,y)y/t^2+2/t;
tspan=[1 5];
y0=1;
[t,y]=ode45(f,tspan,y0)
plot(t,y)
```

```
%Plot the family of the approximation solution of the following initial
%value problem on the interval [-2 2] , y'(x)=y-3x^2+7  , y(0)=1:0.1:4
clc;
clear all;
f=@(x,y)y-3x^2+7;
tspan=[-2 2];
y0=1:0.1:4;
[x,y]=ode45(f,tspan,y0)
plot(x,y)
```

actice6.m ×

```matlab
%Find numerical solution of following equation y'=2t use time interval [0 5] and y(0)=0
clc;
clear all;
f=@(t,y)2*t;
interval=[0 5];
y0=0;
[t,y]=ode45(f,interval,y0)
```

xab.m ×

```
%Create matrix in MATLAB  A=[1 6 5;2 7 3;7 4 9] , B=[2 8 4;1 5 9;6 7 3]
%i) find diagonal matrix of A ii) sum of second row of matrix B iii)
%sum of third column of matrix A
clc;
clear all;
A=[1 6 5;2 7 3;7 4 9]
B=[2 8 4;1 5 9;6 7 3]
diagonal=diag(A)
sumsecondB=sum(B(2,:))
sumthirdA=sum(A(:,3))
```

matrixab.m

```
%Create matrix in MATLAB A=[3/11 2/5 4/7;2/3  4/5 6/5;7/9 8/6 9/4]
%and B=[1/3 8/9 4;2/9 3/7 9/6 ;6 7/4 5/7] i) find inverse of matrix A
%directly using MATLAB command ii) find A*B iii) find sum of elements
%of last row of matrix A
clc;
clear all
A=sym([3/11 2/5 4/7;2/3  4/5 6/5;7/9 8/6 9/4])
B=sym([1/3 8/9 4;2/9 3/7 9/6 ;6 7/4 5/7])
```

matrixab.m*

```
%Create matrix in MATLAB A=[3/11 2/5 4/7;2/3  4/5 6/5;7/9 8/6 9/4]
%and B=[1/3 8/9 4;2/9 3/7 9/6 ;6 7/4 5/7] i) find inverse of matrix A
%directly using MATLAB command ii) find A*B iii) find sum of elements
%of last row of matrix A
clc;
clear all
A=sym([3/11 2/5 4/7;2/3  4/5 6/5;7/9 8/6 9/4])
B=sym([1/3 8/9 4;2/9 3/7 9/6 ;6 7/4 5/7])
inverseA=inve(A)
multiply=A*B
sumthird=sum(A(3,:))
```

matrixab.m

```
%Create matrix in MATLAB  A=[1 6 5;2 7 3;7 4 9] , B=[2 8 4;1 5 9;6 7 3]
%i) Divide matrix A and B and its result store in C matrix
%ii) find transpose of C matrix iii) Find inverse of matrix B
clc
clear all
A=[1 6 5;2 7 3;7 4 9]
B=[2 8 4;1 5 9;6 7 3]
C=A/B
transC=transpose(C)
inverseB=inv(B)
```

```
%Create matrix in MATLAB A=[1 3 4; 9 5 6;1 1 2]; i) find Eigen value and vector
%ii) find reduce echelon form
clc
clear all
A=[1 3 4;9 5 6;1 1 2]
[P D]=eig(A)
reduceechelon=rref(
```

```
rref(A)
rref(A,tol)
More Help...
```

rixab.m* × Untitled* ×

```
% Create function with name "vectorsumprod" in MATLAB which take two vector as
%input u=[1 4 5 7 9] and v=[3  1 9 5 8] function must calculate sum
%and product of tow vectors
function [sum prod]=vectorsumprod(v,u)
clc
clear all
u=[1 4 5 7 9]
v=[3  1 9 5 8]
sum=u+v
prod=u.*v
end
```

vectorsumprod.m × volume_cone.m* × volume_of_cone.m ×

```
%Write the function which compute the volume of cone function must take
%two parameter as input height and radius give volume as output.
%Formula of cone is  v=1/3 πhr^2
function volume=volume_of_cone(height,radius)
clc;
clear all;
h=3;
r=4;
volume=1/3*(pi*h*r^2)
end
```

```
%Solve the system of linear equation
%   2x+y+z=2, -x+y-z=3, x+2y+3z=-10
clc
clear all
eq1='2*x+y+z=2';
eq2='-x+y-z=3';
eq3='x+2*y+3*z=-10';
[x y z]=solve(eq1,eq2,eq3)
```

vectorsumprod.m × volume_cone.m* × volume_of_cone.m* ×

```
clc;
clear all
a=0;
b=2;
n=20;
dx=(b-a)/n;
f=@(x)exp(x^2);
value=0;
for k=1:n
    c=a+k*dx;
    value=value+f(c);
end
R=value*dx
```

ectorsumprod.m × volume_cone.m* × volume_of_cone.m* × trapizodalrule.m ×

```
function area=trapizodalrule(f,a,b,n)
clc;
clear all;
a=0;
b=2;
n=100;
value=0;
dx=(b-a)/n;
f=@(x)exp(-x^2);
avg=(f(b)+f(a))/2;
for k=1:n-1
    c=a+k*dx;
    value=value+f(c)
end
area=(avg+value)*dx
end
```

Editor - C:\Users\Dell Latitude E7250\volume_of_cone.m*

vectorsumprod.m | volume_cone.m* | volume_of_cone.m* | trapizodalrule.m

```
b=2;
n=20;
dx=(b-a)/n;
f=@(x)2+cos(2*sqrt(x));
avg=f(a)+f(b);
value1=0;
value2=0;
for k=1:n-1
    c=a+k*dx;
    if mod(k,2)==0
        value1=value1+2*f(c);
    else
        value2=value2+4*f(c);
    end
end
answer=value1+value2;
area=(avg+answer)*(dx/3)
```

Command Window

```
   -0.9514

>> cos(2*sqrt(2))

ans =

   -0.9514
```

```
clc;
clear all;
a=1;
b=5;
n=100;
dx=(b-a)/n;
f=@(x)sqrt(1+x^3);
value1=0;
value2=0;
avg=f(a)+f(b);
for k=1:n-1
    c=a+k*dx;
    if mod(k,3)==0
        value1=value1+2*f(c);
    else
        value2=value2+3*f(c);
    end
```

Command Window